Introduction to Cyclic Codes

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Cyclic Codes

- Cyclic codes are an important class of linear error-correcting codes with strong algebraic structure and efficient implementation via shift registers.

Definition (Cyclic Code)

A code C of length n over $\mathbb{F}_q$ is cyclic if:

1. C is a linear code
2. For every codeword $(a_0, a_1, \ldots, a_{n-1}) \in C$, its cyclic shift $(a_{n-1}, a_0, \ldots, a_{n-2})$ is also in C

- When studying cyclic codes, we label coordinates as $0, 1, \ldots, n-1$ and associate each vector

$$(a_0, a_1, \ldots, a_{n-1}) \in V(n, q)$$

with the polynomial

$$a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}.$$

Cyclic Codes: Examples

Examples

- The binary code
$$\{000,\ 101,\ 011,\ 110\}$$
is cyclic, since every cyclic shift of a codeword is again in the code.
- The Hamming code $\mathrm{Ham}(3,2)$ is cyclic. Each codeword is obtained as a cyclic shift of its predecessor.
- The binary linear code
$$\{0000,\ 1001,\ 0110,\ 1111\}$$
is not cyclic. However, by interchanging the third and fourth coordinates, it becomes the cyclic code
$$\{0000,\ 1010,\ 0101,\ 1111\}.$$

Polynomials over Finite Fields

Polynomials over $\mathbb{F}_q$

Let $\mathbb{F}_q$ (or simply $\mathbb{F}$) denote a finite field.

- $\mathbb{F}[x]$ denotes the set of polynomials in x with coefficients in $\mathbb{F}$.
- For
$$f(x) = f_0 + f_1 x + \cdots + f_m x^m, \quad f_m \neq 0,$$
the integer m is called the degree of $f(x)$, denoted $\deg f(x)$.
- f_m is the leading coefficient.
- A polynomial is called monic if its leading coefficient is 1.

- Polynomials in $\mathbb{F}[x]$ can be added, subtracted, and multiplied.
- $\mathbb{F}[x]$ is a ring (but not a field), since polynomials of degree > 0 have no multiplicative inverses.
- For $f(x), g(x) \in \mathbb{F}[x]$,
$$\deg(f(x)g(x)) = \deg f(x) + \deg g(x).$$

Division Algorithm for Polynomials

Division Algorithm

For any $a(x), b(x) \in \mathbb{F}[x]$ with $b(x) \neq 0$, there exist unique polynomials $q(x)$ and $r(x)$ such that

$$a(x) = q(x)b(x) + r(x), \quad \deg r(x) < \deg b(x).$$

This is analogous to the division algorithm in $\mathbb{Z}$.

Example

In $\mathbb{F}[x]$, divide $x^3 + x + 1$ by $x^2 + x + 1$:

$$x^3 + x + 1 = (x + 1)(x^2 + x + 1) + x.$$

Here, $q(x) = x + 1$ and $r(x) = x$.

Polynomials Modulo $f(x)$

Congruence Modulo a Polynomial

Let $f(x) \in \mathbb{F}[x]$ be fixed. Two polynomials $g(x), h(x) \in \mathbb{F}[x]$ are said to be congruent modulo $f(x)$, written

$$g(x) \equiv h(x) \pmod{f(x)},$$

if $g(x) - h(x)$ is divisible by $f(x)$.

The Ring $\mathbb{F}[x]/(f(x))$

- By the division algorithm, every $a(x) \in \mathbb{F}[x]$ is congruent modulo $f(x)$ to a unique polynomial $r(x)$ with

$$\deg r(x) < \deg f(x).$$

- $\mathbb{F}[x]/(f(x))$ denotes the set of all such polynomials.
- Addition is ordinary polynomial addition.
- Multiplication is performed modulo $f(x)$.

Example: $\mathbb{F}_2[x]/(x^2+x+1)$

In $\mathbb{F}_2[x]/(x^2+x+1)$,

$$(x+1)^2 = x^2+2x+1 \equiv x \pmod{x^2+x+1}.$$

The elements of the ring are

$$\{0,\ 1,\ x,\ 1+x\}.$$

Every non-zero element has a multiplicative inverse, hence $\mathbb{F}_2[x]/(x^2+x+1)$ is a field of order 4.

Addition $(+)$

$+$	0	1	x	$1+x$
0	0	1	x	$1+x$
1	1	0	$1+x$	x
x	x	$1+x$	0	1
$1+x$	$1+x$	x	1	0

Multiplication $(\cdot)$

$\cdot$	0	1	x	$1+x$
0	0	0	0	0
1	0	1	x	$1+x$
x	0	x	$1+x$	1
$1+x$	0	$1+x$	1	x

Irreducible Polynomials

When is $\mathbb{F}[x]/(f(x))$ a Field?

It is not true that $\mathbb{F}[x]/(f(x))$ is a field for every polynomial $f(x)$.

- For example, $\mathbb{F}[x]/(x^2+1)$ is not a field.
- The key property that makes $\mathbb{F}[x]/(f(x))$ a field is that $f(x)$ is irreducible.

Definition (Reducible / Irreducible)

A polynomial $f(x) \in \mathbb{F}[x]$ is called reducible if

$$f(x) = a(x)b(x),$$

where $a(x), b(x) \in \mathbb{F}[x]$ and

$$\deg a(x) < \deg f(x), \quad \deg b(x) < \deg f(x).$$

If $f(x)$ is not reducible, it is called irreducible.

Factorization in $\mathbb{F}[x]$

Lemma

Let $f(x) \in \mathbb{F}[x]$.

1. $f(x)$ has a linear factor $(x-a)$ if and only if
$$f(a) = 0.$$
2. If $\deg f(x) = 2$ or 3, then $f(x)$ is irreducible if and only if
$$f(a) \neq 0 \quad \text{for all } a \in \mathbb{F}.$$
3. Over any field,
$$x^n - 1 = (x-1)\left(x^{n-1} + x^{n-2} + \cdots + x + 1\right),$$
where the second factor may be further reducible.

Factorization in $\mathbb{F}[x]$ (Continued)

Proof

Let $f(x) \in \mathbb{F}[x]$.

(i) If $f(x) = (x-a)g(x)$, then clearly $f(a) = 0$. Conversely, if $f(a) = 0$, then by the division algorithm:

$$f(x) = q(x)(x-a) + r(x), \quad \deg r(x) < 1.$$

Hence $r(x)$ is a constant, and since $0 = f(a) = r(a)$, we have $r(x) = 0$.

(ii) A polynomial of degree 2 or 3 is reducible *iff* it has at least one linear factor. By (i), the result follows immediately.

(iii) By (i), $x-1$ is a factor of $x^n - 1$. Performing long division of $x^n - 1$ by $x - 1$ gives the other factor:

$$x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

Finite Fields $\mathrm{GF}(p^h)$, $h > 1$

Key Property

The property of a polynomial being irreducible in $\mathbb{F}[x]$ corresponds exactly to the property of a number being prime in $\mathbb{Z}$.

Theorem

The ring $\mathbb{F}[x]/f(x)$ is a field *if and only if* $f(x)$ is irreducible in $\mathbb{F}[x]$.

Remarks:

- Proof is exactly in same the way of proving Z_m is a field if and only if m is prime.
- For any prime p and any positive integer h, there exists an irreducible polynomial over $\mathrm{GF}(p)$ of degree h.
- By Theorem, this guarantees the existence of fields $\mathrm{GF}(p^h)$ for all integers $h \geq 1$.

Back to Cyclic Codes

- We now return to cyclic codes, equipped with polynomial tools.
- The key idea is to represent vectors by polynomials and cyclic shifts by multiplication by x.

Vector–Polynomial Correspondence

A vector $(a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_q^n$ corresponds to

$$a(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}.$$

The Ring $R_n = \mathbb{F}_q[x]/(x^n - 1)$

- In R_n, multiplication is performed modulo $x^n - 1$.
- Since $x^n \equiv 1 \pmod{x^n - 1}$, multiplication by x causes coefficients to wrap around.

Key Observation

Multiplication by x in R_n corresponds exactly to a cyclic shift of a vector.

Algebraic Characterization of Cyclic Codes

Theorem

A subset $C \subseteq R_n$ is a cyclic code if and only if:

1. *C is closed under addition;*
2. *for all $a(x) \in C$ and $r(x) \in R_n$, we have $r(x)a(x) \in C$.*

Interpretation

Cyclic codes are exactly the ideals of the ring R_n.

Why This Works

- Cyclicity means closure under multiplication by x.
- Closure under multiplication by x implies closure under any polynomial in R_n.
- Linearity ensures closure under addition.

Takeaway

Cyclic structure becomes simple algebra.

Generator Polynomial

Theorem

Let C be a nonzero cyclic code in R_n. Then:

1. *There exists a unique monic polynomial $g(x)$ of smallest degree in C.*
2. $C = (g(x)) = \{a(x)g(x) \mid a(x) \in R_n\}$.
3. $g(x)$ *divides* $x^n - 1$.

Properties of the Generator Polynomial

- The generator polynomial completely determines the code.
- If $\deg g(x) = r$, then
$$\dim(C) = n - r.$$
- Every cyclic code corresponds to a monic divisor of $x^n - 1$.

Example: Binary Cyclic Codes of Length 3

Over $\mathbb{F}_2$,

$$x^3 - 1 = (x + 1)(x^2 + x + 1),$$

where both factors are irreducible.

- Each monic divisor gives a cyclic code.
- This produces all binary cyclic codes of length 3.

Check Polynomial

Let $g(x)$ be the generator polynomial of a cyclic code.

Definition

The check polynomial $h(x)$ is defined by

$$x^n - 1 = g(x)h(x).$$

Theorem

A polynomial $c(x) \in R_n$ is a codeword if and only if

$$c(x)h(x) \equiv 0 \pmod{x^n - 1}.$$

Parity-Check Matrix

- The check polynomial encodes all parity conditions.
- Writing the coefficients of
$$h(x),\ xh(x),\ x^2h(x),\dots$$
  yields the rows of a parity-check matrix.

Summary

$g(x)$ generates the code; $h(x)$ checks membership.

Final Takeaway

- Cyclic codes turn coding problems into algebraic ones.
- Their structure is governed by the factorization of $x^n - 1$.
- Generator and check polynomials enable efficient encoding and decoding.

Cyclic codes = structure + efficiency

Thank you!